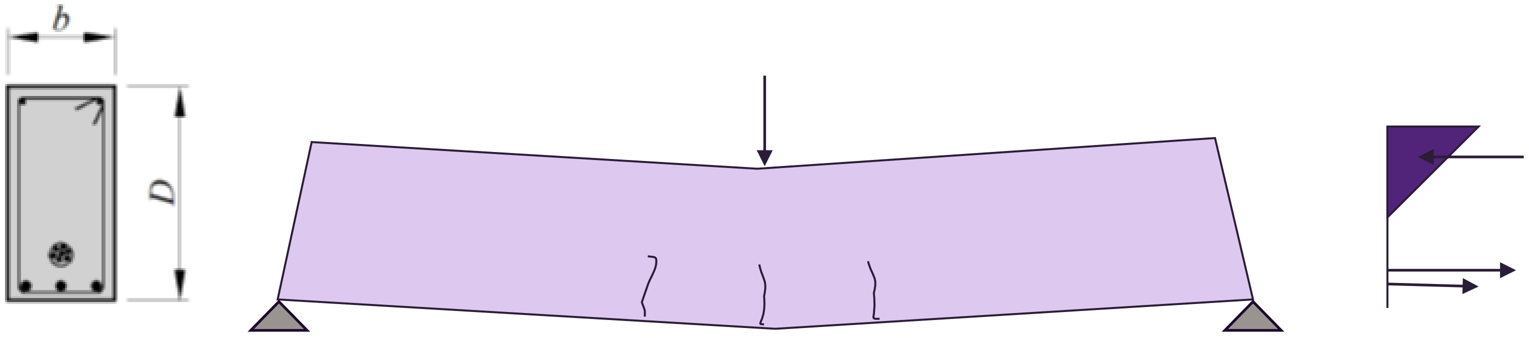
Flexural Behaviour in Post-cracking range



$$I \neq I_g, \quad A \neq A_g$$

Contents

- Re-cap - Flexural behaviour of uncracked beams
- Equivalent Load Concept
- AS3600 method for calculating cracking moment, M_{cr}
- Flexural behaviour of typical cracked prestressed member

•**Chapter 5 Prestressed Concrete, *Warner Foster Gravina* (Reference book)**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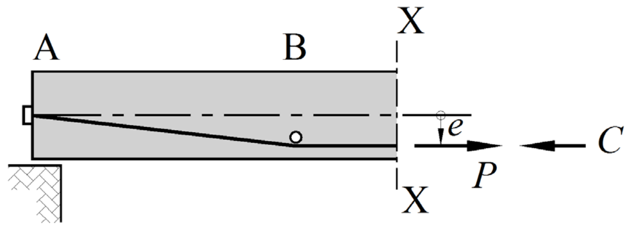
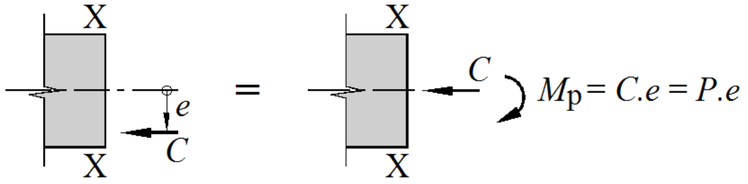
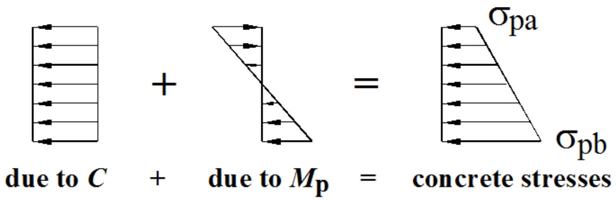
Equilibrating forces at section X-X

Stresses in the concrete induced by the prestress

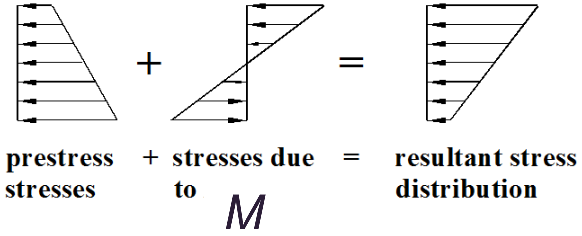




(a) Stress resultants at X-X due to prestress

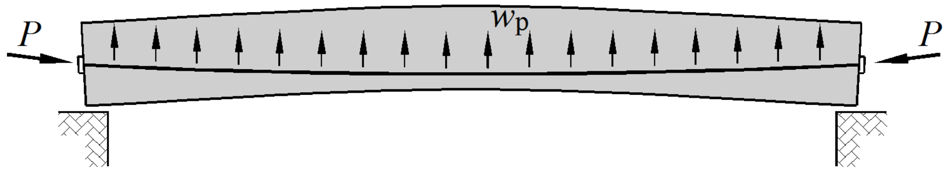


(b) Stresses due to prestress

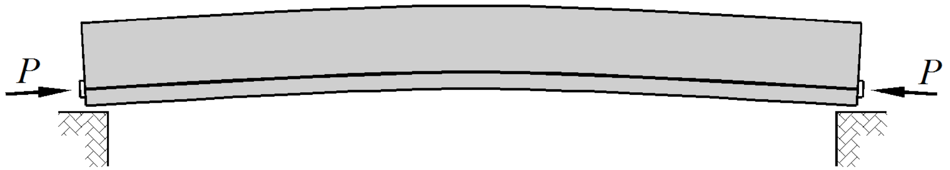


(c) Stresses due to prestress and external load

Parabolic and straight tendons



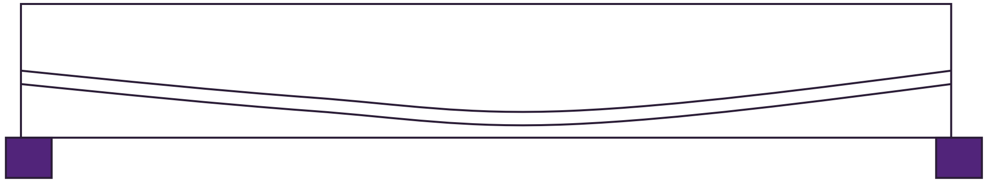
(a) Prestressing with a parabolic tendon



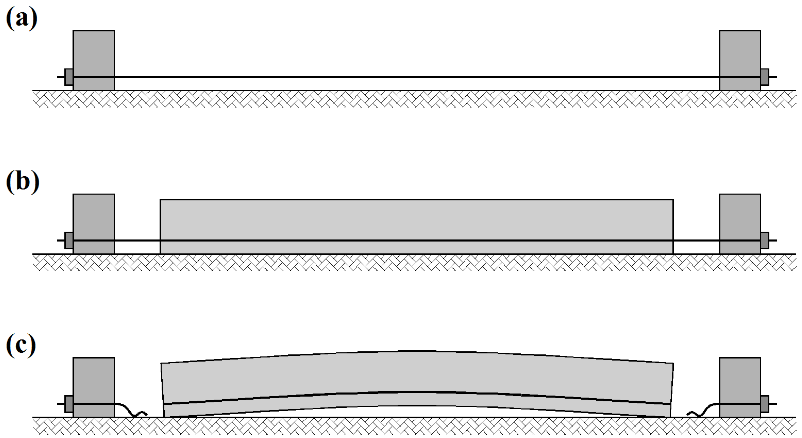
(b) Prestressing with a straight eccentric tendon

Methods of prestressing

Post–tensioning - Beam cast with an unbonded tendon inserted in a duct

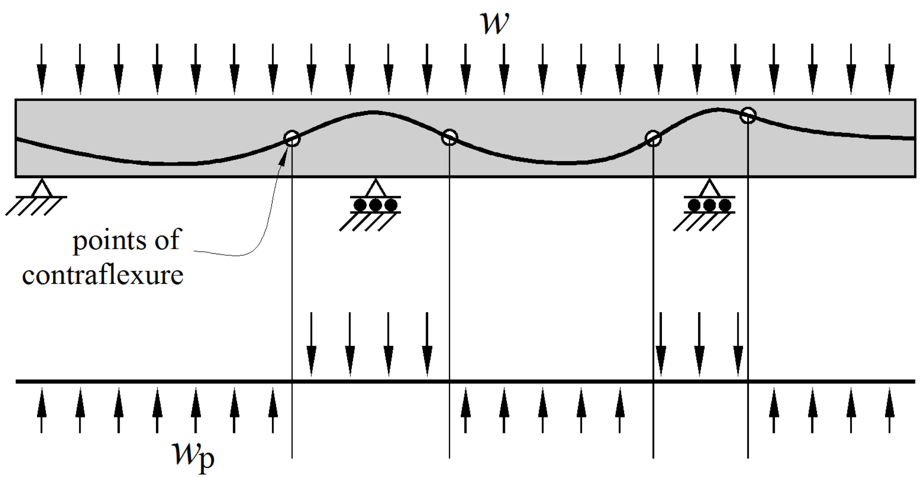


Pre–tensioning – steel is tensioned before concrete is cast



External prestressing - used for repair , external unducted post-tensioned cables or prestressing by external forces

Cable profile and level of prestress



Cable profile continuous member

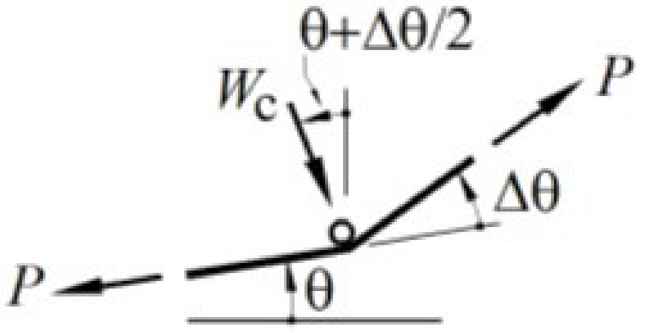
Level of prestress

Fully prestressed – prestress force high enough to eliminate all cracking under service loads

Partially prestressed – precompression not sufficient enough to prevent cracking under full service load. To ensure adequate strength ordinary reinforcement required in tensile regions

Equivalent load at a kink in the tendon

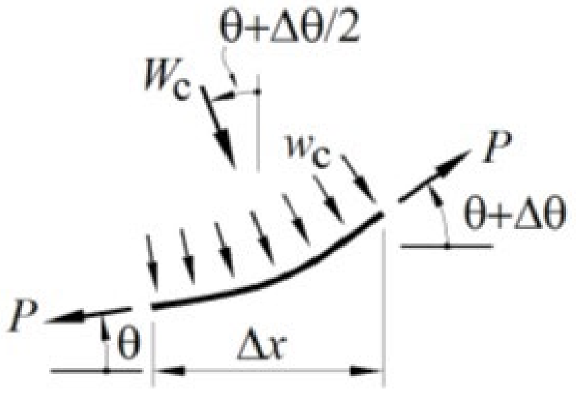
- The slope of the tendon is θ just before the kink and $\theta + \Delta\theta$ just after the kink.
- The tensile force in the tendon on either side of the kink is ***P*** and these are equilibrated by a transverse force $\boldsymbol{W_c}$, which the concrete exerts though the pin on the tendon.
- The tendon, in turn, exerts a force of $\boldsymbol{W_p}$ through the pin on to the concrete which is, of course, equal and opposite to $\boldsymbol{W_c}$.
- For equilibrium in the transverse direction we have a force of $\boldsymbol{W_c} = 2P\sin(\Delta\theta/2)$.
- The angle θ and the kink angle $\Delta\theta$ are small, so that we can set $\sin\theta = \tan\theta = \theta$ and hence write:
- $\boldsymbol{W_p = W_c = P\,\Delta\theta}$
- Can assume that the line of action of the equivalent load ***Wp*** is vertical



(a) Kink in tendon

Equivalent load in a curved region of tendon

- a short length of tendon is curved, and a change in slope $\Delta\theta$ occurs over the length $\Delta\mathbf{x}$.
- The transverse force acting on the concrete is distributed and has the value $\boldsymbol{w_p}$ kN/m, which is, equal and opposite to $\boldsymbol{w_c}$.
- The local stresses are much smaller and there is no danger of local crushing of the concrete.
- The resultant force $\boldsymbol{W_p}$ acting on the concrete over the length Δx is $\boldsymbol{W_p} = \boldsymbol{W_c} = \boldsymbol{w_c}\,\Delta\mathbf{x}$.
- The magnitude of $\boldsymbol{w_p}$ depends on the force $\boldsymbol{P}$ and the curvature κ in the cable, $\boldsymbol{w_p} = \boldsymbol{P} \times \kappa$. If the tendon has a parabolic shape then $\boldsymbol{w_p}$ is uniformly distributed. .

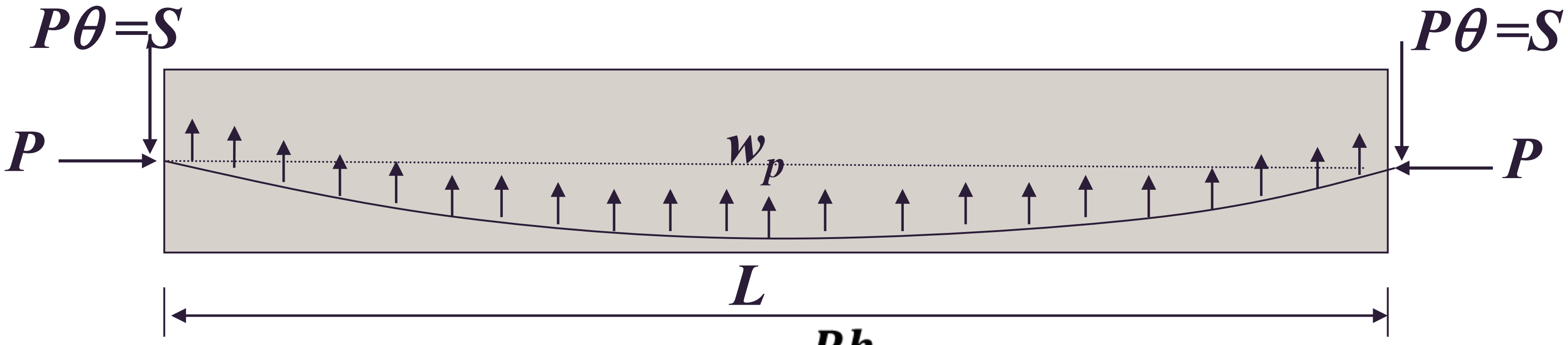


(b) Curve in tendon

Equivalent load –Parabolic cable

The slope θ of the tendon at the end of the beam (at the anchorage) is 4h/L,

Transverse end force exerted through the anchorage is ***S = 4Ph/L***, while the horizontal force is taken to be approximately ***P***.



$$w_p = -8\frac{Ph}{L^2}$$

$$Slope\,\theta(x) = \frac{de}{dx} = \frac{4h}{L}\left(1 - \frac{2x}{L}\right)$$

$$Slope\,\theta(x = 0\ and\ x = L) = \frac{4h}{L}$$

Cracking moment, Clause 8.5.3.1 of AS3600

Applied moment at which the bottom fibre of the beam reaches the tensile cracking stress of concrete

$$M_{cr} = Z(f'_{ct.f} - \sigma_{cs} + P/A_g) + Pe \geq 0$$

Z = section modulus uncracked section I_g/y

σ_{cs} = maximum shrinkage induced tensile stress on uncracked section at the extreme fibre at which cracking occurs taken as

$$\sigma_{cs} = E_s \varepsilon^*_{cs}(2.5p_w - 0.8p_{cw})/(1+50p_w)$$

Web reinforcement ratio for tensile reinforcement

$$p_w = (A_{st} + A_p)/(b_w d)$$

Web reinforcement ratio for compressive reinforcement

$$p_{cw} = A_{sc}/b_w d$$

ε^*_{cs} final design shrinkage strain of concrete

$f'_{cf.f}$ = flexural tensile strength of concrete

$$f'_{ct.f} = 0.6\sqrt{f'_c}$$

Calculation cracking moment

Calculate the cracking moment and the cracking load for the beam given in week 1 lecture (Example 4.1).
The beam has a rectangular cross-section 800 mm deep and 400 mm wide has a span of 10 metres.
It is post-tensioned by a cable that has a cross-sectional area of 1000 mm^2 and an eccentricity *e* that varies parabolically from zero at the ends to a maximum of *e* = 250 mm at midspan, where the effective prestressing force is 1200 kN.

Other Data:
$f'_{cf.f}=0.6\ Sqrt(35) = 3.55MPa$, $A_g = 320x10^3mm^2$, $I_g = 17.07\ x10^9mm^4$,
$Z_{bot}=Z_{top}=I_g(D/2) = 17.07x10^9/400 = 42.67x10^6mm^3$

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Residual shrinkage-induced stress:

For this section we have:

$$p_w = \frac{1000}{400 \times 650} = 0.00385 \quad p_{cw} = 0$$

and the residual stress is calculated from Equation 5.1 as:

$$\sigma_{cs} = \frac{200\times10^3 \times 565\times10^{-6} \times 2.5 \times 0.00385}{1 + 50 \times 0.00385} = 0.91 \text{ MPa}$$

Cracking moment:

The average prestress on the section is:

$$\frac{P_e}{A_g} = \frac{1200\times10^3}{320\times10^3} = 3.75 \text{ MPa}$$

and the cracking moment is (Equation 5.4):

$$\begin{aligned} M_{cr} &= Z(f'_{ct.f} - \sigma_{cs} + P_e/A_g) + P_e e \\ &= 42.67\times10^6(3.55 - 0.91 + 3.75) + 1200\times10^3 \times 250 \\ &= 273.7\times10^6 + 300.0\times10^6 \\ &= 574\times10^6 \text{ Nmm} \quad (574 \text{ kNm}) \end{aligned}$$

Cracking load:

For a uniformly distributed load, the load to cause cracking is:

$$w_{cr} = \frac{8M_{cr}}{L^2} = \frac{8 \times 574}{10^2} = 45.9 \text{ kN/m}$$

From Example 4.1 the self weight was $w_G = 8.0$ kN/m and, thus, the additional uniformly applied load that can be carried above the self weight, without cracking, is $45.9 - 8 = 37.9$ kN/m.

Cracking moment for prestressed section with reinforcement

Recalculate the cracking moment for a similar section to the one considered previously except that in addition t the tendon it contains tensile reinforcing steel of 4N24 bats for which A_{st} = 1800mm²

Using a slightly larger effective depth of $d = 700$ mm to allow for the equivalent depth of the combined steel area (tendon and reinforcing steel), we have:

$$p_w = \frac{1000 + 1800}{400 \times 700} = 0.0100; \quad p_{cw} = 0$$

and the shrinkage induced stress is calculated from Equation 5.1 as:

$$\sigma_{cs} = \frac{200 \times 10^3 \times 565 \times 10^{-6} \times 2.5 \times 0.0100}{1 + 50 \times 0.0100} = 1.88 \text{ MPa}$$

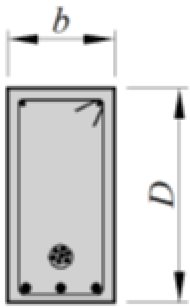
and thus the cracking moment is (Equation 5.4):

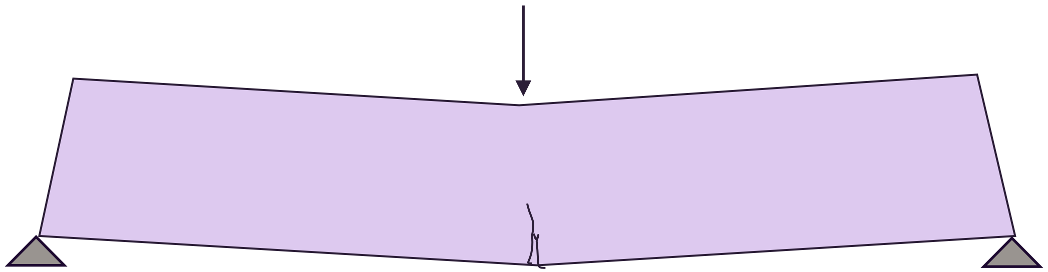
$$\begin{aligned} M_{cr} &= Z(f'_{ct.f} - \sigma_{cs} + P_e/A_g) + P_e e \\ &= 42.67 \times 10^6 (3.55 - 1.88 + 3.75) + 1200 \times 10^3 \times 250 \\ &= 231.3 \times 10^6 + 300.0 \times 10^6 \\ &= 531 \times 10^6 \text{ Nmm} \quad (531 \text{ kNm}) \end{aligned}$$

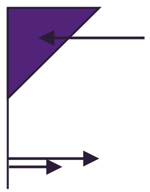
Compared to previous example cracking moment has been reduced substantially as the results of the restraint shrinkage provided by the tensile reinforcement

Post-cracking flexural behaviour

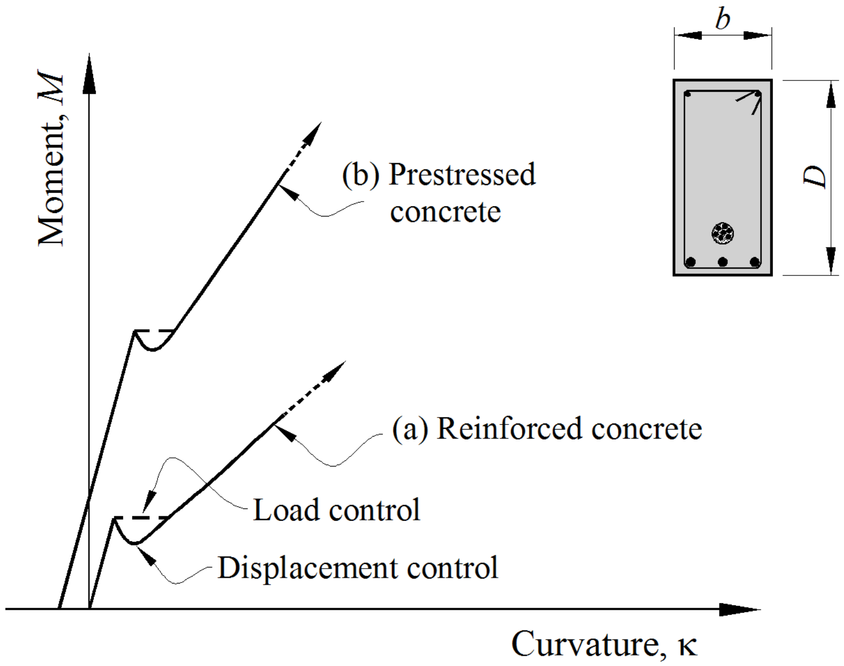
We now examine the post-cracking flexural behaviour in the region of the beam surrounding the section of maximum moment when the first crack has formed.







Transition from uncracked to cracked conditions



- Sudden loss in concrete area in tensile zone of RC or PC section at first cracking
- Section loses flexural stiffness during changeover from uncracked to cracked section
- Presence of eccentric prestress in section provides for an initial negative curvature

Post cracking behaviour

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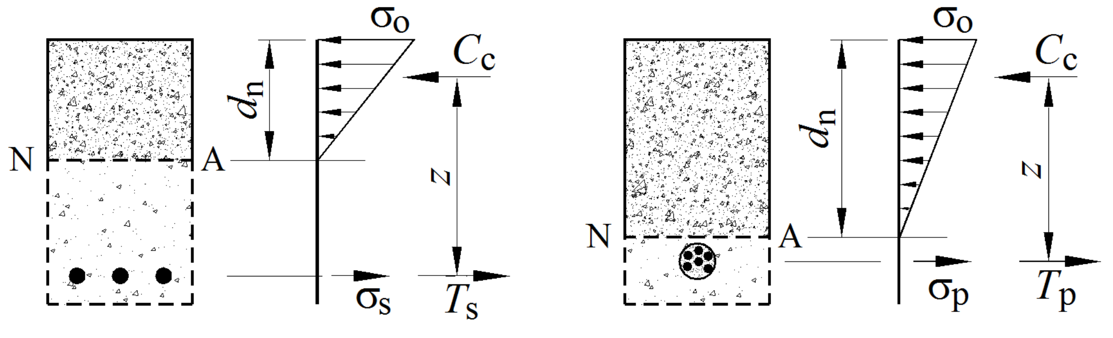
Post cracking stress distribution of a section

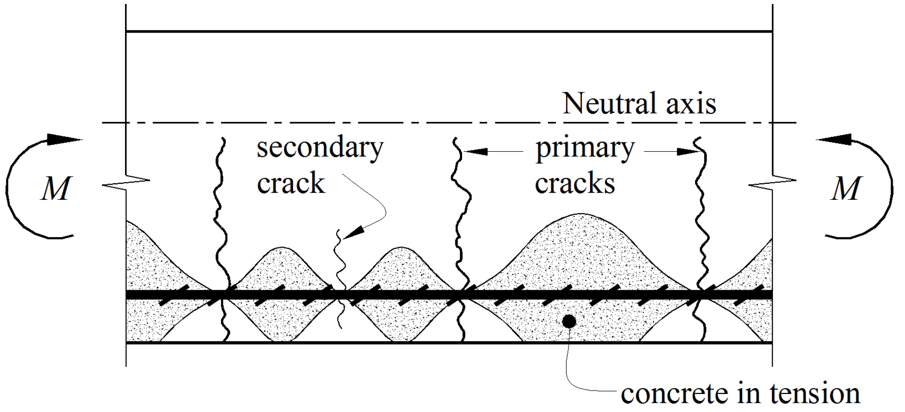


(a) Reinforced concrete (d_n, z constant)

(b) Prestressed concrete (d_n, z vary with M)

- Level of prestress effects post cracking behaviour
- N/A position and crack tip rise with increase in applied moment

- Flexural cracks and tension stiffening regions



Partially Prestressed Beams

In all our previous calculations on prestressed concrete beams we've used the Gross Cross Sectional area and Gross second moment of area of the section

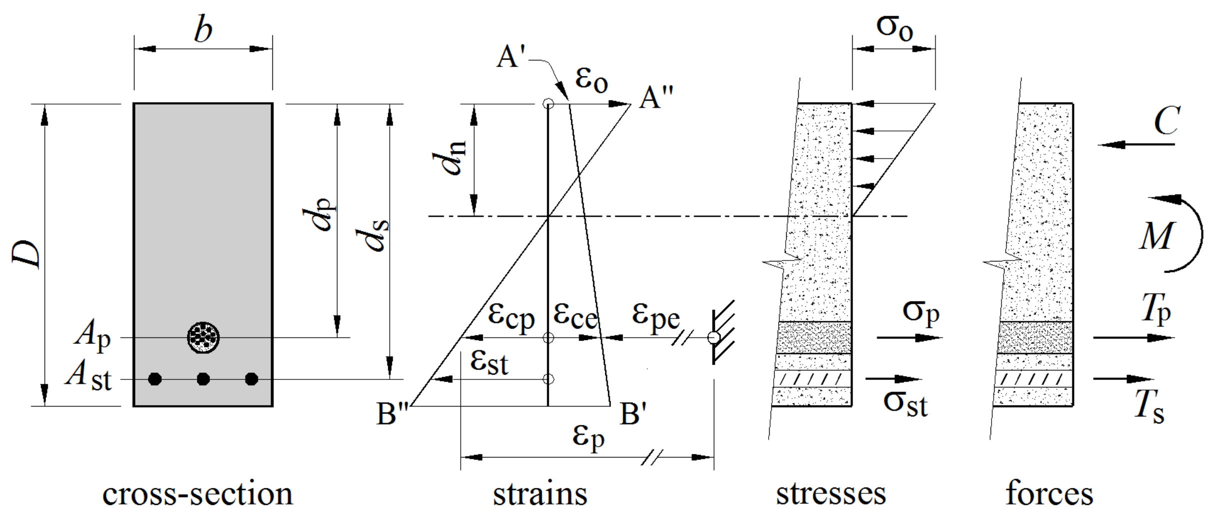
In doing that we assume that the cross-section is uncracked (Tension at any point in the section doesn't exceed cracking stress of concrete)

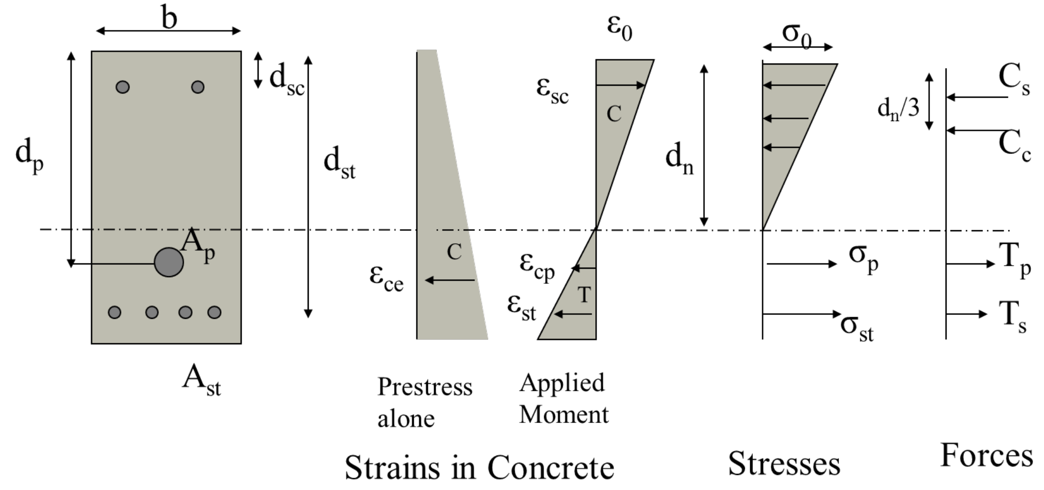
In normal designs, we can allow prestressed concrete beams to be cracked at service loads, although not as severely as in reinforced concrete

Beams which are designed to crack at service loads are known as "partially prestressed beams".

Most prestressed beams are designed as "Partially prestressed beams"

Elastic Analysis of a rectangular cracked section



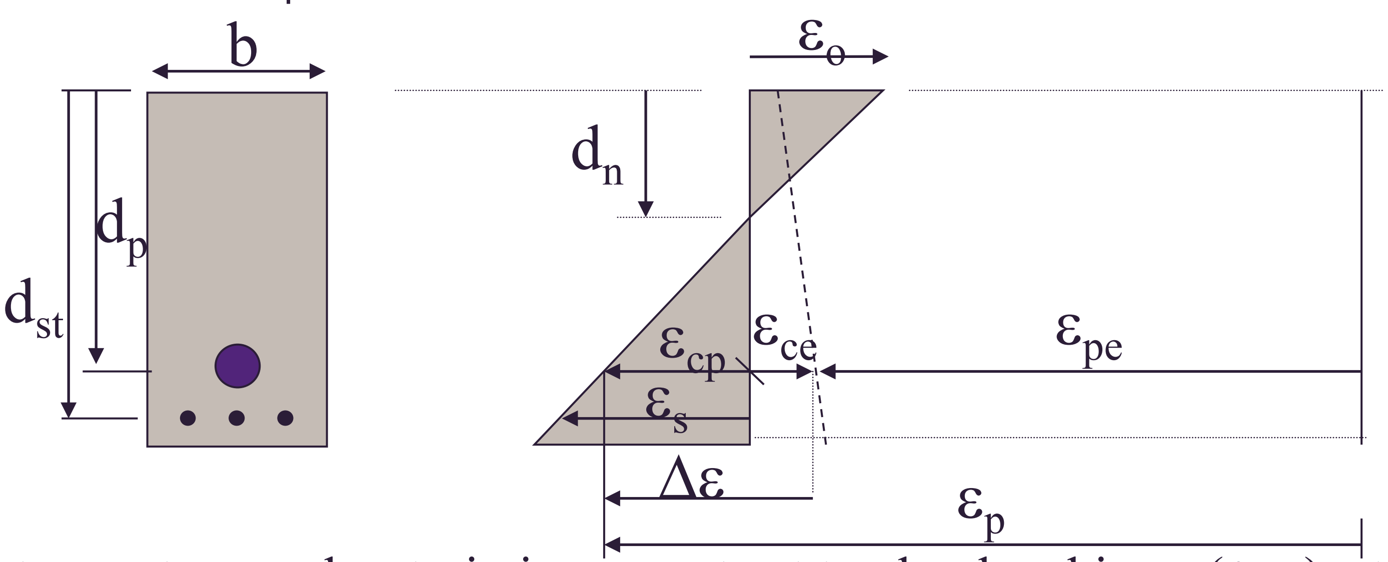


Elastic analysis for a prestressed beam section differs from that for a reinforced concrete section in one key step

This is the step in which the strain in the prestressing steel is related to that in the adjacent concrete

Determining strain in prestress steel

Strains in section with prestress



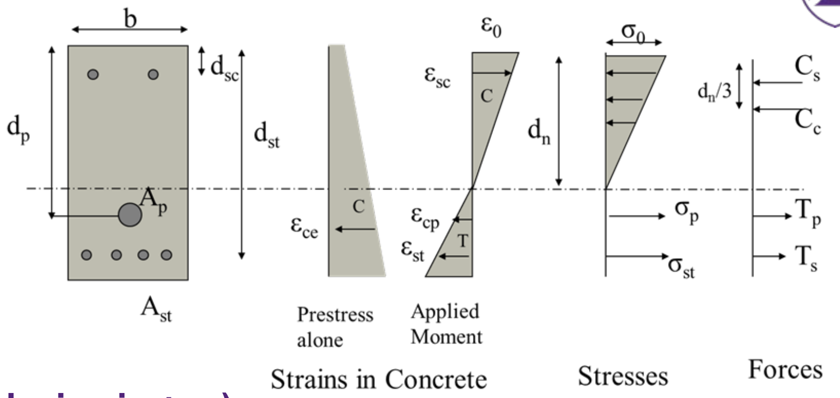
Due to prestress only, strain in concrete at tendon level is ε_{ce} (+ve); strain in the tendon is ε_{pe} *(-ve)*

Applied moment causes strain in concrete to change from +ve ε_{ce} to –ve ε_{cp}. The concrete at level of the tendon has therefore undergone a tensile strain increment $\Delta\varepsilon = \varepsilon_{cp} + \varepsilon_{ce}$ (absolute values). Assuming perfect bond, the same change in strain must occur in tendon.

Total strain in prestress $\varepsilon_p = \varepsilon_{pe} + \varepsilon_{ce} + \varepsilon_{cp}$

Where $\varepsilon_{cp} = \varepsilon_0(d_p - d_n)/d_n$

Equations of equilibrium for the section



Force Equilibrium

$$C_c + C_s = T_p + T_s$$

Moment Equilibrium,

Taking moments about the top of the section, (anti-clockwise is +ve)

$$M = T_s d_{st} + T_p d_p - C_c d_n/3 - C_s d_{sc}$$

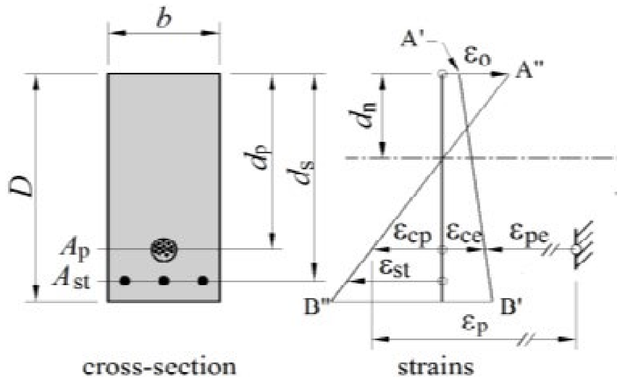
(line of action C_c is $d_c = d_n/3$)

Where 'M' is the applied external moment

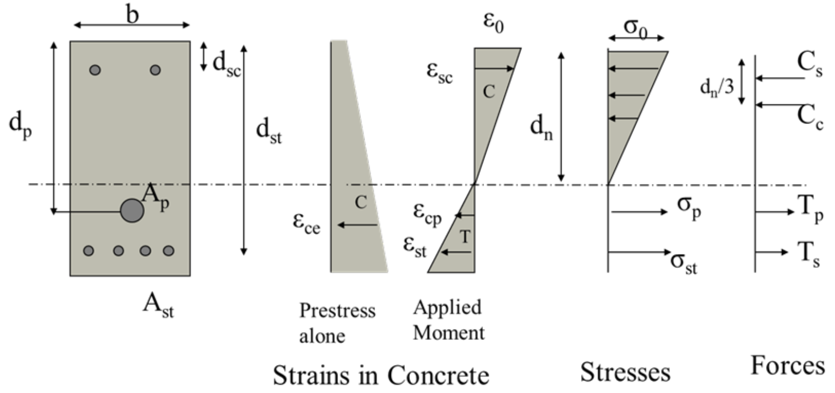
Tensile strain in the concrete at the levels of the reinforcement and prestressing steel

$$\varepsilon_{st} = \varepsilon_{cst} = \varepsilon_0 (d_{st} - d_n)/d_n$$

$$\varepsilon_{cp} = \varepsilon_0 (d_p - d_n)/d_n$$



Calculation of forces in the section



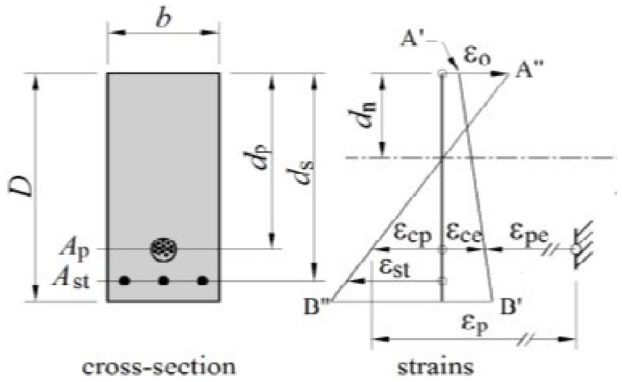
$$C_c = 0.5\sigma_0 b d_n = 0.5E_c\varepsilon_0 b d_n$$

$$C_s = A_{sc}E_s\varepsilon_{sc} = A_{sc}E_s\,\varepsilon_0\,(d_n-d_{sc})/d_n$$

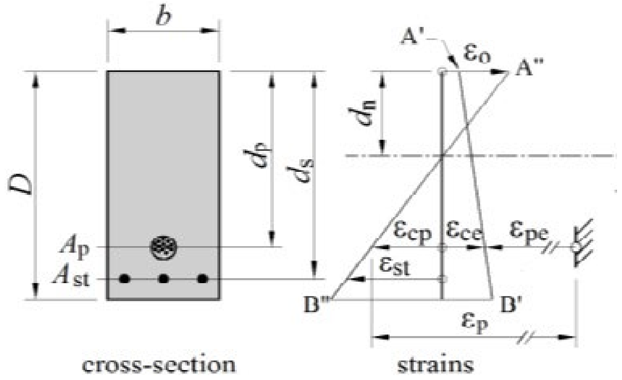
$$T_s = A_{st}\sigma_{st} = A_{st}E_s\,\varepsilon_{st} = A_{sc}E_s\,\varepsilon_0\,(d_{st}-d_n)/d_n$$

$$T_p = A_p\sigma_p = A_pE_p\,\varepsilon_p$$

For a given strain diagram of concrete, all the parameters can be established. Only tricky issue is establishing ε_p.



Strain in prestressing steel



due to effective prestress

$\varepsilon_{pe} = \sigma_p/E_p = P/(A_pE_p)$

Compressive stress in concrete adjacent to the prestressing cable, due to prestress alone (use $P = P_e$ = effective prestress force)

$\varepsilon_{ce} = \sigma_{ce}/E_c$ **where** $\sigma_{ce} = P/A_g + Pe^2/I_g$

Tensile strain in concrete at level of prestress steel due to applied Moment

$\varepsilon_{cp} = \varepsilon_0(d_p - d_n)/d_n$

Total strain in prestress steel = $\varepsilon_p = \varepsilon_{pe} + \varepsilon_{ce} + \varepsilon_0(d_p - d_n)/d_n$

Force Equilibrium

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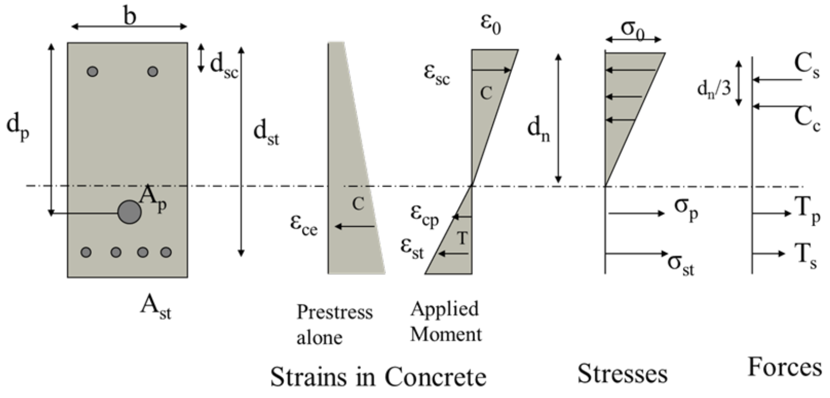
$C_c + C_s = T_p + T_s$

$0.5 E_c \varepsilon_0 b d_n + A_{sc} E_s \, \varepsilon_0 \, (d_n - d_{sc})/d_n =$

$A_{sc} E_s \, \varepsilon_0 \, (d_{st} - d_n)/d_n + A_p E_p \, \varepsilon_p$



Since ε_p is a function of d_n

ε_0 and d_n are the unknowns in the equation.

We need another equation to solve for the two unknowns.

Moment Equilibrium

$M = T_s d_{st} + T_p d_p - C_c d_n/3 - C_s d_{sc}$

$M = A_{sc} E_s\, \varepsilon_0\, (d_{st} - d_n)/d_n \times d_{st} + A_p E_p\, \varepsilon_p \times d_p$

$\quad - 0.5 E_c \varepsilon_0 b d_n^2/3 - A_{sc} E_s\, \varepsilon_0\, (d_n - d_{sc})/d_n \times d_{sc}$

The curvature in the section when the moment acts is

$$\kappa = \varepsilon_o/d_n$$

Most of the time, what we need are the stresses in the section for a given service moment. Therefore the moment is known. Again we have the two unknowns: ε_0 and d_n

Now, you can solve for the two unknowns. We usually adopt a trial and error procedure, ie assume a d_n and calculate ε_0 using force equilibrium and check the moment.

To determine moment curvature relationship for a general section

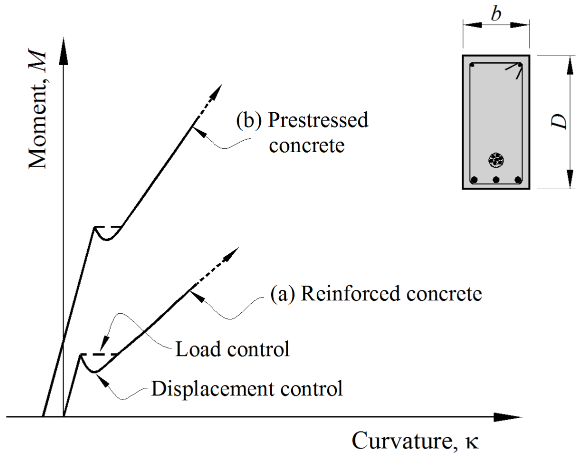
Choose progressively decreasing sequence of values of d_n that begin at value slightly smaller than d_p

For a chosen d_n find corresponding top fibre strain e_o, use trail and error to satisfy force equilibrium. $C_c+C_s = T_p + T_s$

With values of e_o and d_n evaluate stresses and strains in the section

Evaluate moment from $M= T_s d_{st} + T_p d_p - C_c d_n/3 - C_s d_{sc}$ that corresponds to chosen value of d_n and calculate section curvature $k = e_o/d_n$

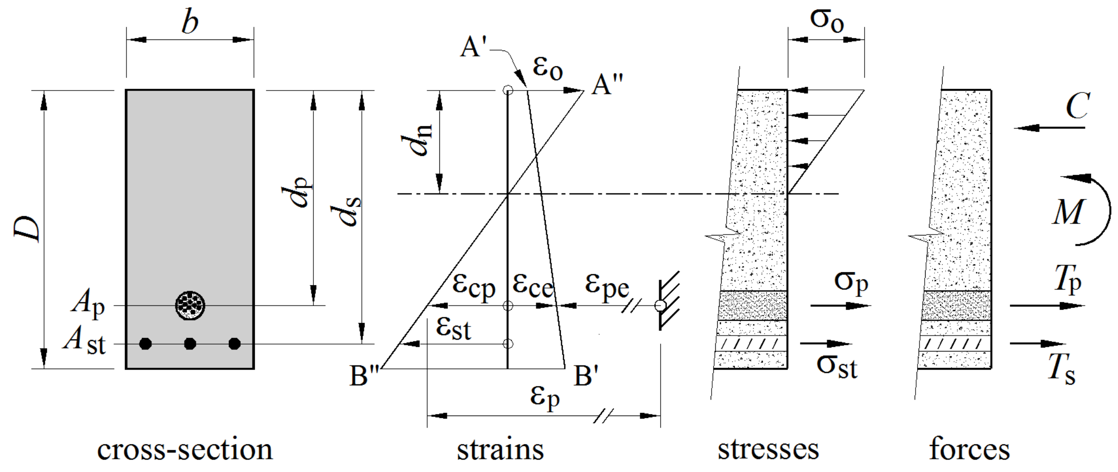
Repeat the calculations for other values of d_n



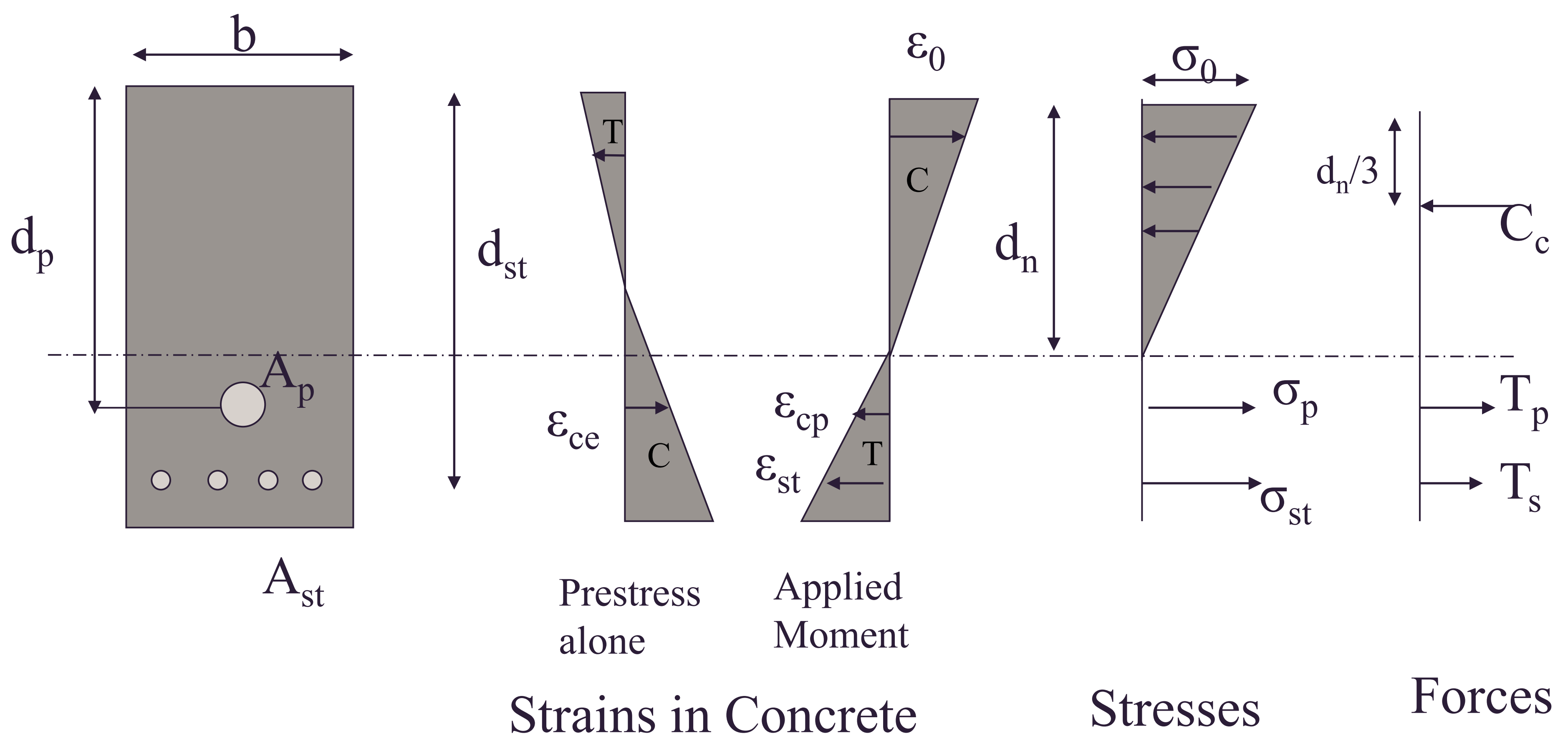
Simple rectangular section Closed form Algebraic expression for top fibre strain ε_o

(no compression steel)

$$\varepsilon_o = \frac{A_p E_p d_n (\varepsilon_{ce} + \varepsilon_{pe})}{0.5 E_c b d^2{}_n - E_s A_{st}(d_s - d_n) - E_p A_p (d_p - d_n)}$$



Elastic Analysis of a rectangular cracked section



Worked example- Calculation of stress in a prestressed concrete section under a given bending moment

A post-tensioned prestressed concrete beam has a configuration of reinforcement shown in the figure. Calculate the maximum stress in concrete at a service bending moment of 345 kNm.

f'_c = 40 MPa

E_c = 32000 MPa

A_{st} = 1350 mm^2

f_{sy} = 400 MPa

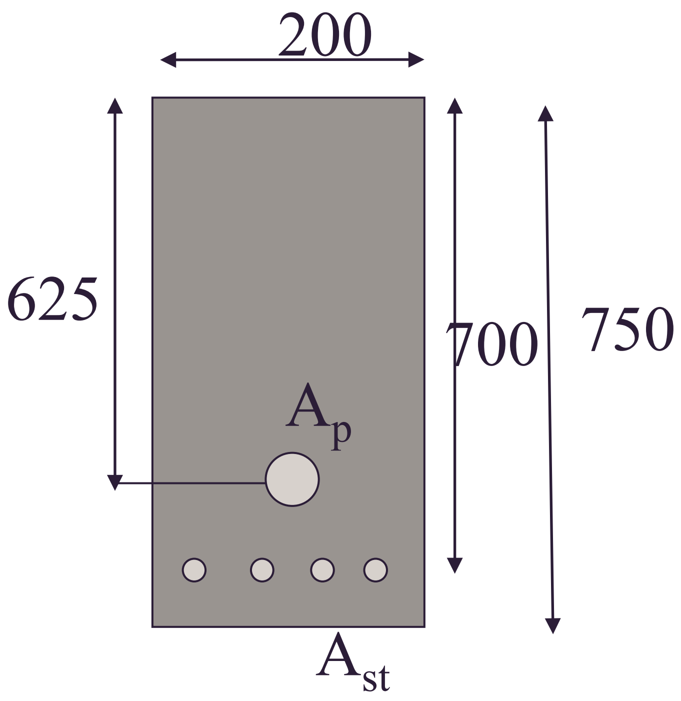
A_p = 405 mm^2

f_{py} = 1710 MPa

ε_{py}= 0.009

Effective Prestress force, P_e is 445.5 kN after all losses.

M_{cr} = 238kNm, section cracked



Let's start with ε_p.

$$\varepsilon_{pe} = \sigma_p/E_p = P/AE_p =$$

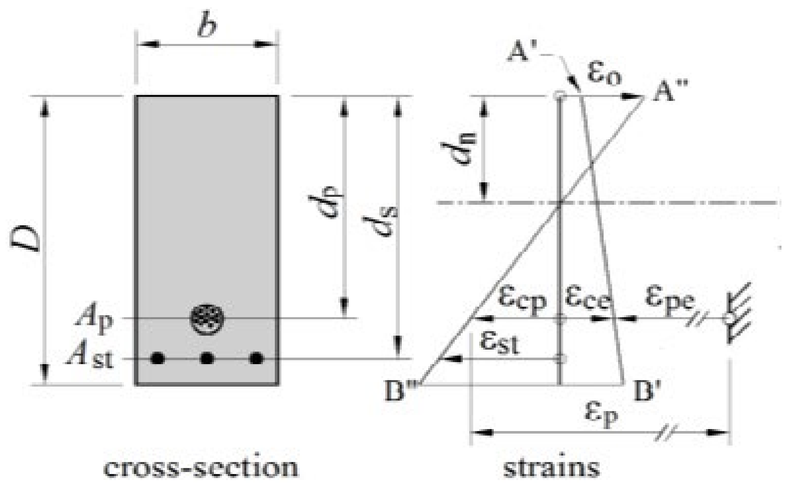
$$=445000/(405\text{x } 190000)=0.00579$$

$$\varepsilon_{ce} = (P/A_g + Pe^2/I_g)/E_c = 6.93/32000$$

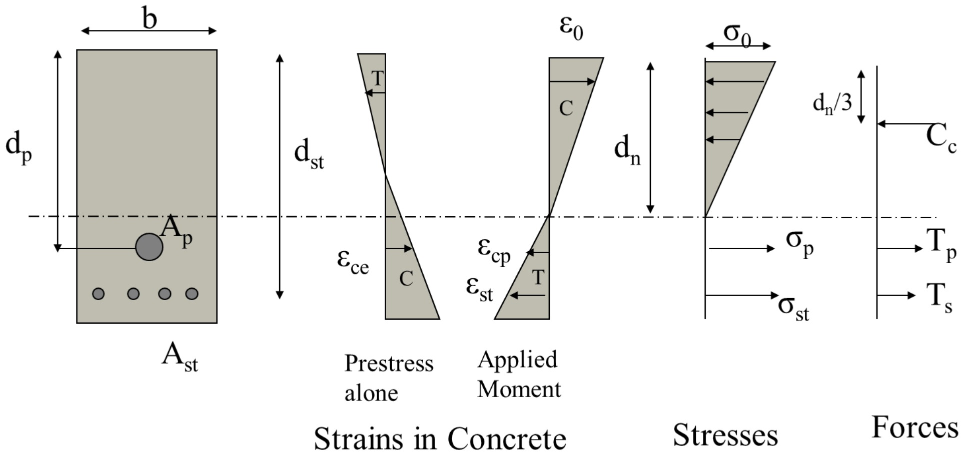
$$= 0.000217$$

$$\varepsilon_{cp} = \varepsilon_0(d_p\text{-}d_n)/d_n = \varepsilon_0(625\text{-}d_n)/d_n$$

ε_p = 0.00579 + 0.000217 + $\varepsilon_0(625\text{-}d_n)/d_n$



Force Equilibrium



$C_c = T_p + T_s$

$0.5E_c\varepsilon_0 b d_n = A_{st}E_s\,\varepsilon_0\,(d_{st}-d_n)/d_n + A_pE_p\,\varepsilon_p$

(note no compression steel in this problem)

0.5x 32000 x ε_0 x200d_n = 1350 x 200000 x ε_0 (700-d_n)/d_n + 405 x 190000 x (0.00579 + 0.000217 + ε_0(625-d_n)/d_n))

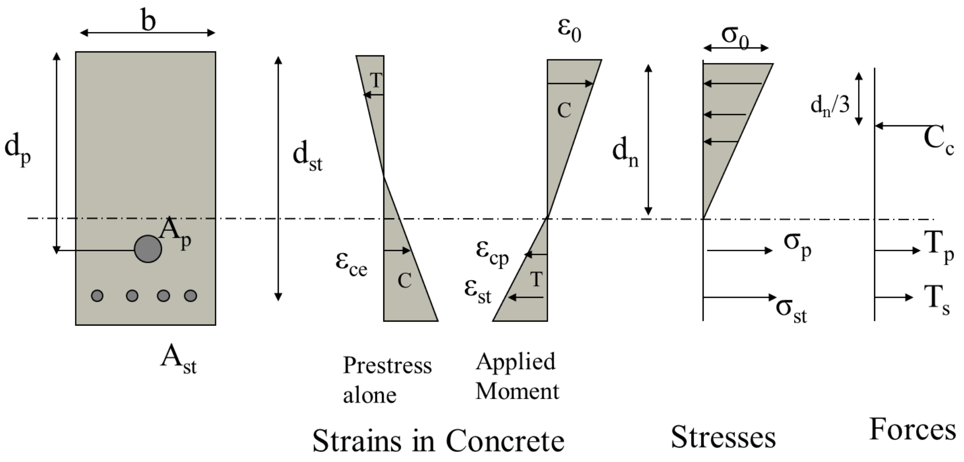
ε_0(0.5x 32000 x 200d_n - 1350 x 200000(700-d_n)/d_n - 405 x 190000 (625-d_n)/d_n) = 405 x 190000 x (0.00579 + 0.000217)

ε_0=405 x 190000 x (0.00579 + .000217)/

(0.5x 32000 x 200d_n - 1350 x 200000(700-d_n)/d_n - 405 x 190000 (625-d_n)/d_n)

(Remember $\varepsilon_{st} = \varepsilon_0(d_{st}-d_n)/d_n$ and $\varepsilon_{cp} = \varepsilon_0(d_p-d_n)/d_n$)

Trial and error procedure



Assume d_n = 375 mm

ε_0= 0.505 x 10^{-3}

σ_0 = 32,000 x 0.505 x 10^{-3} = 16.2 MPa

C_c = 0.5x 32000 x 0.505 x 10^{-3} x200x 375 =606 kN

$\sigma_{st=} E_s \varepsilon_0 (d_{st}-d_n)/d_n$ = 200000 x 0.505 x 10^{-3} (700-375)/375 = 87.5 MPa

T_s = 87.5 x 1350 = 118 kN

$T_{p\,=} A_pE_p\, \varepsilon_p$ = 405 x 190000 x (0.00579 + 0.000217 + 0.505 x 10^{-3}(625-375)/375 = 488 kN

$M = \ T_sd_{st} + T_pd_p - C_cd_n/3$

= 118 x 700 + 488 x 625 – 606 x 375/3 = 312 kNm

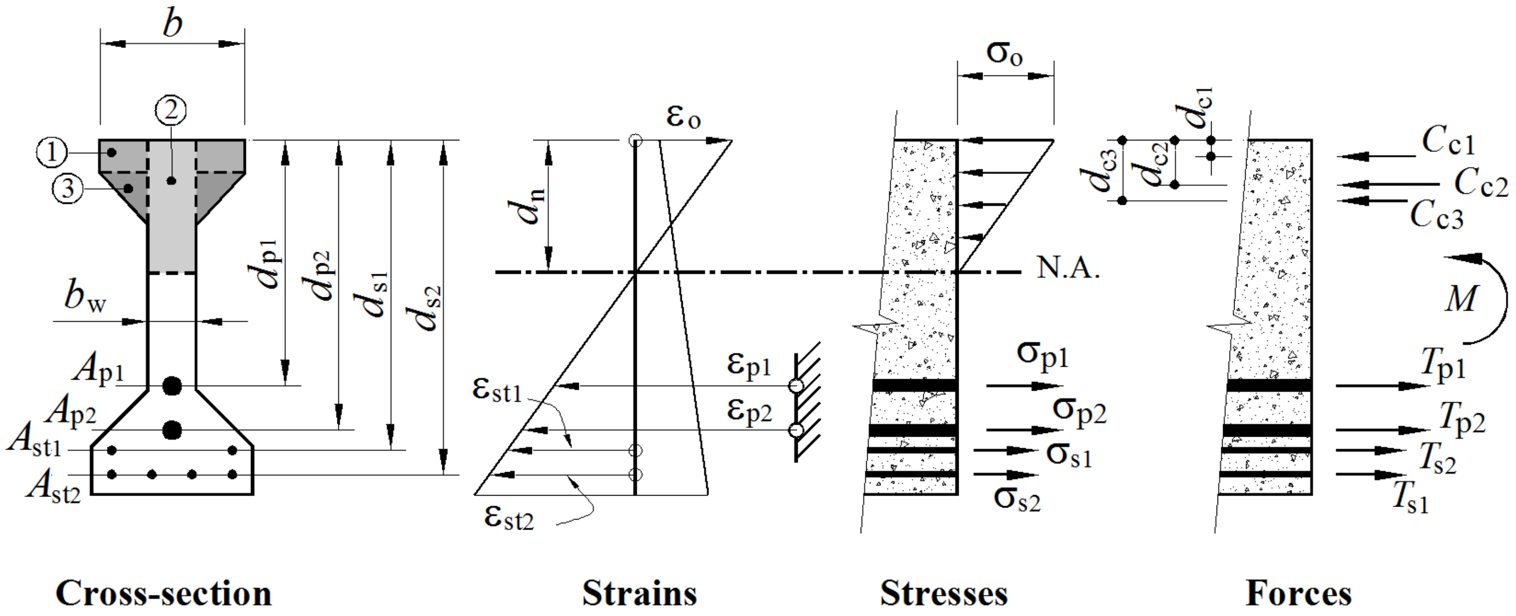
We need the stresses and strains at a moment of 345 kNm.

Should we increase the d_n or should we decrease it ?

Relationship between applied moment and d_n

d_n mm	ε_0	σ_0 MPa	C kN	Ts kN	Tp kN	M kNm
375	.000505	16.2	606	118	488	312
350	586	18.7	656	117	158	345
325	703	22.5	731	165	219	512
400	447	14.3	572	91	482	288

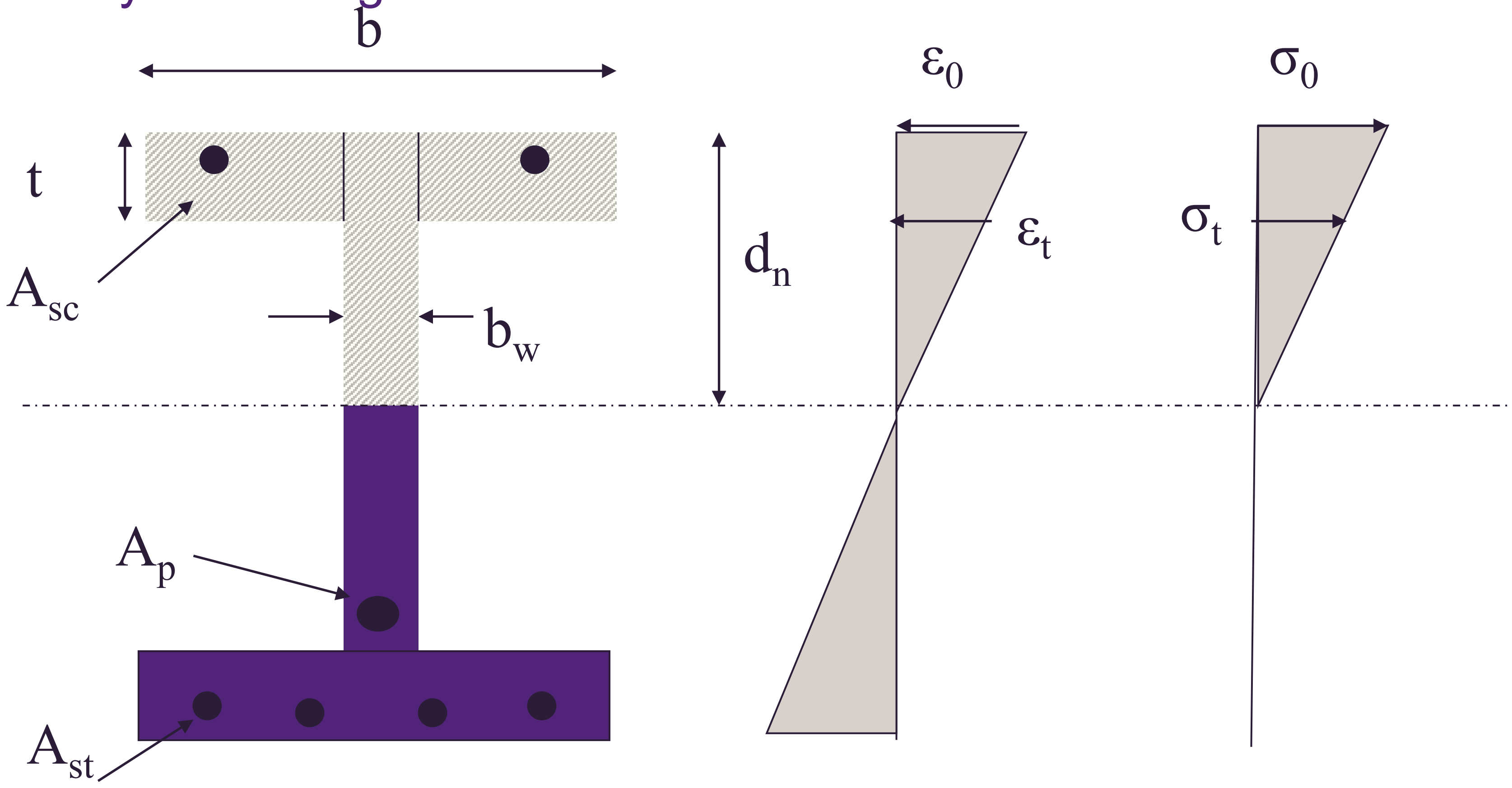
Cracked section analysis: general trial and error method



$$\sum_i C_i = \sum_k T_{pk} + \sum_j T_{sj}$$

$$M = \sum_k T_{pk}.d_{pk} + \sum_j T_{sj}.d_{sj} - \sum_i C_i.d_{ci}$$

Cracked section analysis of flanged beams



Strain and stress at bottom of flange

$\varepsilon_t = \varepsilon_0(d_n\text{-}t)/d_n, \quad \sigma_t = E_c\varepsilon_0(d_n\text{-}t)/d_n$

Equation for the compressive forces change as follows

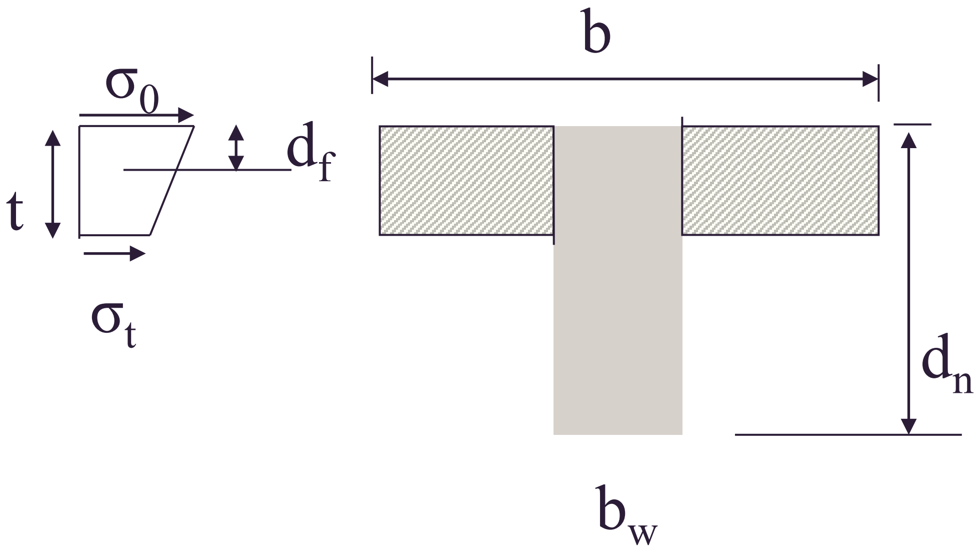
Force in flange outstands

$C_f = 0.5(\sigma_0 + \sigma_t)(b-b_w)t$

$$C_f = 0.5 \times \left[E_c \varepsilon_0 + E_c \varepsilon_0 \frac{(d_n - t)}{d_n} \right] (b - b_w) t$$

This will act at a distance from the top of

$d_f = t(3d_n-2t)/(6d_n-3t)$



Force in web

$C_w = 0.5\ \sigma_0\ b_w d_n$

$$C_w = 0.5 \times E_c \varepsilon_0 \times b_w \times d_n$$

Web force will act at depth $d_n/3$

Force in top compressive reinforcement

$$C_s = A_{sc}E_s \times \frac{\varepsilon_0}{d_n}(d_n - d_{sc})$$

Force in bottom tensile reinforcement

$$T_s = E_s A_{st}\varepsilon_0 \times \left(\frac{d_{st} - d_n}{d_n}\right)$$

Force in cables

$$T_p = E_p A_p \times (\varepsilon_{ce} + \varepsilon_{pe} + \varepsilon_{cp})$$

Consider force equilibrium :

$$(C_f + C_w) + C_s = T_s + T_p$$

$$\left(0.5 \times (E_c\sigma_0 + E_c\varepsilon_0 \frac{d_n - t}{d_n})(b - b_w)t + 0.5E_c\sigma_0 b_w d_n\right) + A_{sc}E_s \times \frac{\varepsilon_0}{d_n}(d_n - d_{sc}) = E_s A_{st}\varepsilon_0 \times \left(\frac{d_{st} - d_n}{d_n}\right) + E_p A_p \times (\varepsilon_{ce} + \varepsilon_{pe} + \varepsilon_{cp})$$

re - arranging gives

$$\varepsilon_0 = \frac{E_p A_p d_n(\varepsilon_{ce} + \varepsilon_{pe})}{0.5E_c b_w d_n{}^2 + E_c(d - 0.5t)(b - b_w)t + E_s A_{sc}(d_n - d_{sc}) - E_s A_{st}(d_{st} - d_n) - E_p A_p(d_p - d_n)}$$

Moment Equilibrium gives

Taking moments about the top fibre of the section

$$M = T_s d_{st} + T_p d_p - C_w d_n/3 - C_f d_f - C_s d_{sc}$$

This analysis assumes that the neutral axis is in the web of the beam. A check needs to be performed to ensure that it is the case.

When do you need a cracked section analysis ?

To determine stresses at the extreme compression fibre of concrete σ_0 to ensure that stress is within acceptable limits

We have done this for an uncracked section in previous weeks, which is quite straightforward when the section is uncracked

To determine the cracked second moment of area I_{cr} of the section to then calculate the effective second moment of area I_{eff} for calculation of deflections

Serviceability limit state design of PC members – crack control

<u>Design at transfer</u> P_i. AS3600 Clause 8.1.6.2

Only immediate losses have occurred. Member is subject to the minimum applied loading ie self-weight and prestress only.

Minimum allowable concrete strength at transfer = f_{cp}

Determines the upper limit of the magnitude of the prestressing force

AS3600 Clause 8.1.6.2 provides a 'deemed to satisfy' criterion if the maximum compressive stress in the concrete at transfer is limited to

- For a section that is rectangular in cross section and where stress distribution is triangular in shape, $0.6f'_{cp}$ *; otherwise*
- $0.5f'_{cp}$ where f'_{cp} is the characteristic strength of concrete at transfer

Serviceability limit state design of PC members – crack control (cont'd)

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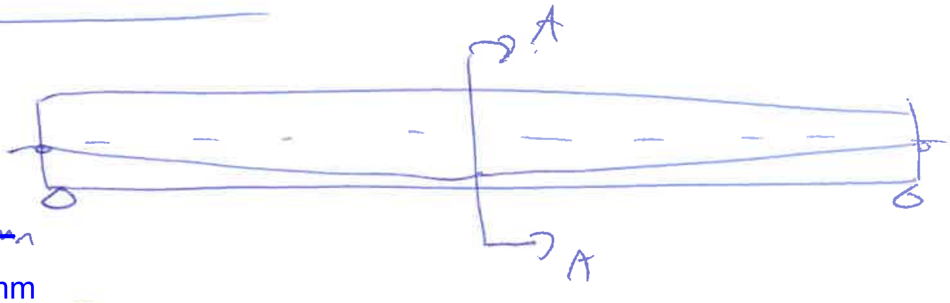
<u>Design for stress limit under full service load</u>

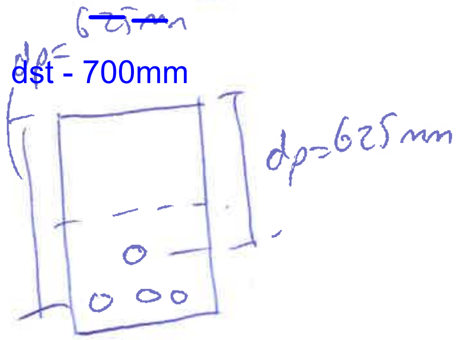
Fully prestressed members are required to remain uncracked under full service load, and so maximum tensile stresses ≤ 0.25 $\sqrt{f'_c}$ Clause 8.6.3

For partially PC members, some cracking is permitted and Cl 8.6.3 requires:

- reinforcement to be provided near the tensile face
- the calculated maximum flexural tensile stress under short term service loads to ≤ 0.6 $\sqrt{f'_c}$ to control cracking in a 'deemed to satisfy' provision

Lecture Problem – wk3





e @ mid span $= 625 - \frac{750}{2} = \underline{250\text{ mm}}$

@ A A

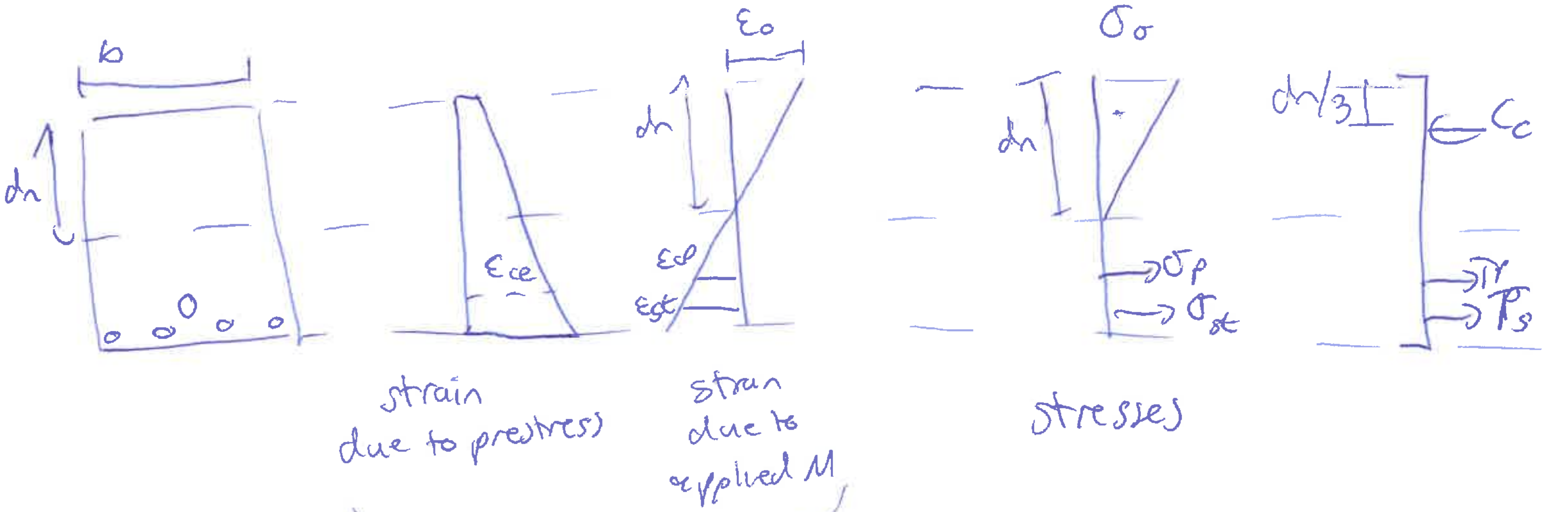
M_s = service bending moment
$= 345$ kNm

P_e = effective Prestress force = 446 kN. after all losses

$M_{cr} = 238$ kNm   since $M_s > M_{cr}$

$\therefore$ section is cracked.

$\Rightarrow$ Calculate stress in concrete in PC member due to $M_s = 345$ kNm



strains in concrete

ε_p = total strain in prestress

$$= \underbrace{\varepsilon_{pe}}_{\text{strain in prestress steel}} + \underbrace{\varepsilon_{ce} + \varepsilon_{cp}}_{\text{strain in concrete @ level of prestress}}$$

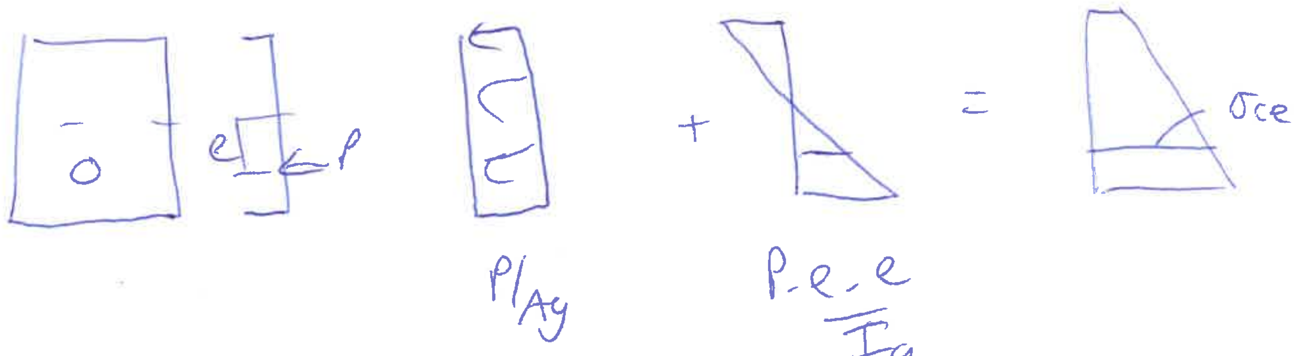
ε_{pe} = strain in prestress steel due to applied prestress force

$$= \frac{\sigma_p}{E_p} = \frac{\left(P_e/A_p\right)}{E_p} = \frac{\left(\frac{446\times10^3}{405}\right)}{190{,}000}$$

$$= \underline{0.006}$$

ε_{ce} = strain in concrete at level of prestress steel due to prestress force alone.

$$\sigma_{ce} = \frac{P}{A_g} + \frac{(P.e.).e}{I_g}$$



$$\sigma_{ce} = \frac{446\times10^3}{150{,}000} + \frac{446\times10^3 \times 250\times250}{\left(\frac{200\times750^3}{12}\right)}$$

$= 6.9$ MPa.

$$\varepsilon_{ce} = \frac{\sigma_{ce}}{E_c} = \frac{6.9}{32{,}000} = \underline{0.0002}$$

ε_{cp} = strain in concrete @ level of prestress steel due to applied moment

$$= \frac{\varepsilon_o\,(d_p - d_n)}{d_n} = \frac{\varepsilon_o\,(625-d_n)}{d_n}.$$

2

$$\varepsilon_p = \varepsilon_{pe} + \varepsilon_{ce} + \varepsilon_{cp}$$

$$\varepsilon_p = 0.006 + 0.0002 + \frac{\varepsilon_o (625 - d_n)}{d_n}$$

Force equilibrium

$$C_c = T_p + T_s$$

$$0.5\, \varepsilon_o E_c \cdot b \cdot d_n = A_p \cdot \varepsilon_p \cdot E_p + A_{st} \cdot E_{st} \cdot \frac{\varepsilon_o (d_{st} - d_n)}{d_n}$$

$$0.5 \times \varepsilon_o \times 32{,}000 \times 200 \times d_n = 405 \times 190{,}000 \times \left[0.0062 + \frac{\varepsilon_o (625 - d_n)}{d_n} \right]$$

$$+ 1350 \times 200{,}000 \times \frac{\varepsilon_o (700 - d_n)}{d_n}$$

$$\varepsilon_o = \frac{(405 \times 190{,}000 \times 0.0062)}{\left[0.5 \times 32{,}000 \times 200 \times d_n\right] - \left[1350 \times 200{,}000 \frac{(700 - d_n)}{d_n}\right] - \left[405 \times 190{,}000 \times \frac{(625 - d_n)}{d_n}\right]} \quad (1)$$

trial + error procedure

Assume $d_n = 375\text{ mm}$ sub into equation (1)

$\varepsilon_o = 0.505 \times 10^{-3}$

$\sigma_o = \varepsilon_o E_c = 0.505 \times 10^{-3} \times 32{,}000 = 16\text{ MPa}.$

$\Rightarrow C_c = 0.5 \times \sigma_o \times b \cdot d_n$

$= 0.5 \times 16 \times 200 \times 375$

$= 606{,}000\text{ N}$

$\sigma_{st} = E_s \cdot \frac{\varepsilon_o (d_{st} - d_n)}{d_n} = 200{,}000 \times 0.505 \times 10^{-3} \times \frac{(700 - 375)}{375}$

$= 87.5 \text{ MPa}$

$\Rightarrow T_{st} = A_{st} \cdot \sigma_{st} = 1350 \times 87.5 = 118{,}000 \text{ N}$

$\Rightarrow T_p = A_p \cdot E_p \cdot \varepsilon_p = 405 \times 190{,}000 \times 0.0062 + 0.505 \times 10^{-3} \times \frac{(625 - 375)}{375}$.

$T_p = 488{,}000 \text{ N}$

$\Rightarrow$ Calculate internal moment.

$M = T_{st} \cdot d_{st} + T_p \cdot d_{pt} - C_c \frac{d_n}{3}$

$= 118{,}000 \times 700 + 488{,}000 \times 625 - 606{,}000 \times 375/3$

$M = 312 \times 10^6 \text{ Nmm} = 312 \text{ kNm} < (M_s\ 345)$

we need σ_o & ε_o @ moment of 345 kNm

if $d_n = 325$ mm $\quad \varepsilon_o = 0.703 \times 10^{-3} \quad M = 512$ kNm

if $d_n = 350$ mm $\quad \varepsilon_o = 0.59 \times 10^{-3} \quad M = 345$ kNm

$\sigma_o = 22.5 \text{ MPa}$